Statement of Work (SOW)
NASA Goddard Space Flight Center (GSFC) Earth Sciences Division
Snow4Flow Earth Venture Suborbital

Background:
The purpose of this SOW is to describe the firm fixed price work to be performed by the Contractor to modify and operate an aircraft used for the measurement of seasonal snow depth and glacier thickness. This work will support the Snow4Flow Earth Venture Suborbital mission. Snow4Flow is a NASA airborne mission to improve our ability to interpret ongoing and future changes in Northern Hemisphere glaciers.

Scope of Work:
The Contractor shall perform the firm fixed price work necessary to support instrumentation integration and test flights for Snow4Flow. The Contractor shall provide the necessary qualified personnel, materials, equipment, documentation, and facilities (unless otherwise stated) to perform the work.

Period of Performance:
Engineering: 01 June 2026 – 16 August 2026
Integration: 17 August 2026 – 29 August 2026
Test flights: 30 August 2026 – 05 September 2026
Deinstall: 06 September 2026 – 12 September 2026

Project Specific Requirements:
The Contractor shall perform the following:

- Provide the Basler BT-67 with tail number C-FDXQ. Due to the nadir port configurations required by the Snow4Flow instrument suite, this specific aircraft must be used. If C-FDXQ is unavailable, then another Basler BT-67 suitable for this mission shall be selected by the vendor in concurrence with the NASA Science Point of Contact (POC). In such a case, the vendor will work with the Snow4Flow Science Team to ensure that the instrument suite can be installed in the replacement aircraft.
- Perform the engineering and aircraft modifications to fly the following instrument packages provided by the Snow4Flow Science Team. NASA Science POC will provide detailed specifications and requirements for each instrument package. Any required fabrication must be completed prior to the integration start date listed in the Period of Performance section.
    - Arizona Radio Echo Sounder (ARES), University of Arizona
    - CReSIS UWB Snow Radar, University of Kansas
    - Riegl VQ-580II-S, University of Alaska
    - RGB Camera, NASA Goddard Space Flight Center
- Provide instrument racks and accessories for integration of Snow4Flow equipment.
- Provide a minimum of 6 instrument operator seats.
- Assist with instrument engineering and integration.
- Provide designated Snow4Flow Science Team members access to Contractor facilities, including but not limited to, hanger space housing C-FDXQ.

- Provide NASA with the necessary determinations of compliance with FAA/TCCA airworthiness standards (14 CFR Part 23 or 14 CFR Part 25 or Canadian equivalent depending on the type of plane selected).
- Satisfy the NASA airworthiness review process as identified in NAII 7900.3, *Airworthiness Review Process for the Eastern Region Airworthiness Review Board (ER-ARB)* for aircraft operations conducted in the performance of this activity, including but not limited to: participating in meetings and reviews; providing documentation of work performed, responding to any action items from the review board, etc.
    - Upon award of contract, begin participation in NASA airworthiness review process – review dates are tentatively scheduled for:
        - Final Airworthiness Review: 13 August 2026
        - Operational Readiness Review: 20 August 2026
    - Milestones will be determined by the airworthiness review schedule in consultation with the Contractor.
    - Support any site inspections that may be required as part of the NASA airworthiness process.
    - The Contractor along with the Science POC shall present at the Operational Readiness Review (ORR). Any actions to the Contractor from this review shall be completed prior to the issuance of a Flight Release. All ORR material shall comply with NPR 7900.3, *Aircraft Operations Management*, Section 3.12.
    - If the aircraft identified for this work is replaced by another aircraft during or after completion of any part of the airworthiness or readiness review process, then parts or entire reviews may need to be repeated for the new aircraft.
- Carry out science flights as specified below. Dates specified in Period of Performance section. Total science and test flight hours: up to 12.
    1. Local test flight over water in the Great Lakes region or off the east coast of North America.
    2. Optional second local test flight depending on resulting data quality.
- Estimated proposal of costs shall be submitted to the Contracting Officer Representative (COR) and Contracting Officer (CO) for review and approval. Work cannot proceed without approval from the COR and CO.
- The Contractor shall provide the COR and CO aircraft metrics by 30 days after the end of the period of performance, to include aircraft type, total flight hours, number of sorties flown, sorties aborted due to aircraft issues and sorties aborted not due to aircraft issues. This information will be used by the Government to complete the mandatory Government Services Administration (GSA) Federal Aviation Interactive Reporting System (FAIRS) report for GSFC aviation operations.
- The Contractor shall provide the Science POC flight times for each flight conducted under this SOW at the conclusion of each flight. Flight times shall be determined engine on to engines off. The Science POC will use these times to submit daily flight reports to the NASA Airborne Science Program which are due prior to the next scheduled flight.
- All Contractor flight crew and Qualified Non-Crewmembers (QNC) shall comply with OCHMO-STD-1880.1, *NASA Aviation Medical Certification Standards*. QNCs shall also complete required NASA QNC forms as well as perform any QNC training as required.

NASA Commercial Aviation Services Information (Required per NAII 7900.3):

- **Aviation services** contracted for exclusive use fall under Commercial Aviation Services (CAS) and are considered a Governmental Aircraft under General Services Administration (GSA) rule. All government aircraft in support of NASA will be overseen by NASA during the exclusive use contract. If a US Operator is selected, this will normally fall under the definition of 'Public Use' per Federal Aviation Administration (FAA) Advisory Circular AC 00-1.1B *Public Aircraft Operations – Manned and Unmanned*. If outside the US, the operator will comply with both NASA and the home country's aviation regulations and airworthiness.

- **'Public Use' operations** as defined in FAA Advisory Circular 00-1.1B is an aircraft used only for the US Government, is not performing an operation with a commercial purpose, and has only the necessary crew onboard. For the duration of flight testing required by this SOW, the selected aircraft and flights meet the FAA definition of 'public use', therefore the US Government (NASA in this case) is required to conduct a review of the company, an airworthiness review of the aircraft, and a review the proposed flight test in lieu of the respective FAA review. All three reviews must be complete, and approvals issued prior to first flight, and the continuous surveillance component continues throughout the entire review, approval, and flight test campaign.

- **NASA is required to conduct these reviews** whether the company is a US company or not, and whether the flight operations are in the US or not. If a prime contractor elects to sub-contract flight activity, the prime contractor shall facilitate NASA conducting these reviews with authorities appropriate for that company and that country.

- **Commercial Air Services (CAS)**. For crewed aircraft, NASA HQ will conduct a review of the selected company to ascertain they meet the requirements to provide CAS aircraft and to conduct flight operations to the US Government. Following the HQ CAS review, the GSFC Chief of Flight Operation will review results and determine any mitigations required to conduct the proposed test. If a Contractor has not completed a previous NASA HQ site visit inspection, then the site inspection shall occur prior to the award of a contract for NASA aviation services.

- **Airworthiness**. NASA's Eastern Region Airworthiness Review Board (ER-ARB) will conduct a review of the aircraft and all proposed modifications. Any modifications to the aircraft that deviate from an FAA (or civil aviation authority if overseas) approved airworthiness configuration shall be reviewed and approved by the ER-ARB. The ER-ARB does have requirements and a suggested format to present material, however with prior coordination, the ER-ARB is willing to accept the Contractor's existing process and documentation. If the Contractor's normal airworthiness process is used, NASA requests the ability to participate in the review, and to be provided documentation of their meeting minutes, action items, and close out of those items. At the conclusion of the NASA airworthiness review, a NASA Statement of Airworthiness will be issued.

- **Flight Release**. The GSFC Chief of Flights Operations or designee will conduct a review of the proposed flight test operation. This review includes the flight test plan, test points, flight procedures, airspace, and crew qualifications. At the end of this review, the Chief of Flight Operations will issue a Flight Release. Following issuance of the Flight Release an Approval to

Proceed Review with GSFC senior management occurs where a final approval will be granted for the start of all flight operations.

- **Continuous Surveillance**. The GSFC Chief of Flight Operations is required to provide continuous surveillance of any contracted NASA flight activity. The results of the reviews above will determine the level and scope of this continuous surveillance plan. Range of scope could include on site oversight of planned maintenance and flight activities.

Deliverables:
Any associated documentation, science data, hardware, etc. collected; generated or manufactured/purchased in the execution of this SOW shall be the sole property of NASA at the conclusion of the contract.

Final Closeout:
This SOW will be considered completed when the Contractor's aircraft returns to the Contractor's facility, all Snow4Flow equipment has been removed, and all deliverables have been provided to NASA.

Points of Contact:
[redacted]